

Reflective Portfolio Report

Ethical Analysis of the Epic Sepsis Model: An AI-Enabled Clinical Decision Support System

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Chapter 1: Introduction

Sepsis is a life-threatening condition caused by a dysregulated host response to infection and is associated with high mortality and substantial healthcare costs. In hospitalised patients, sepsis occurs in approximately 2–6% of admissions, with reported mortality rates ranging from 15% to 40% depending on severity and timeliness of intervention [1].

Clinical evidence consistently shows that earlier identification and treatment—particularly timely administration of antibiotics significantly reduce mortality. As a result, healthcare systems have increasingly adopted AI-enabled Clinical Decision Support Systems (CDSS) to assist clinicians in early detection and risk stratification of sepsis [2].

This project focuses on a real-world, deployed AI-enabled CDSS: the **Epic Sepsis Model (ESM) Inpatient Predictive Analytic Tool**, integrated into the Epic electronic health record (EHR) system. Epic Systems is the largest EHR provider in the United States, with systems serving approximately 54% of patients nationwide, making ESM one of the most widely deployed AI-based decision support tools in clinical practice [1].

The ESM is designed to continuously monitor hospitalised patients and generate a numerical risk score indicating the likelihood of sepsis, enabling earlier clinical intervention. Unlike experimental or research-only models, the ESM operates in live hospital environments and directly influences clinical workflows. Its outputs are used to trigger alerts to nurses and physicians when a predefined risk threshold is exceeded, thereby shaping diagnostic attention, treatment prioritisation, and escalation decisions. As such, the ESM represents a safety-critical AI system whose ethical implications extend beyond model accuracy to issues of trust, accountability, over-reliance and human–AI collaboration [3].

Key Features of the System

The Epic Sepsis Model is a data-driven, sub-symbolic AI system implemented within a commercial electronic health record platform. It is proprietary; however, published validation studies provide sufficient evidence to characterise its technical structure and behaviour [1].

AI Technology and Model Characteristics

- **Model type:** Data-driven predictive model (sub-symbolic)
- **Input features:** Approximately 80 clinical and demographic variables, including vital signs, laboratory results, comorbidities, and patient demographics
- **Training data:** Developed using approximately 500,000 patient encounters
- **Output:** A continuously updated numerical sepsis risk score
- **Deployment:** Fully integrated into the Epic EHR, operating in real time on hospitalised patients

The ESM does not rely on explicit clinical rules, symbolic reasoning, or hand-crafted knowledge representations. Instead, patient data are combined and weighted according to learned statistical

relationships to produce a risk score. While the system can surface which variables contributed most to an elevated score, its internal decision logic is not fully transparent to end users, placing it firmly in the category of sub-symbolic, black-box clinical decision support [4].

Operational Threshold and Performance

In the validation study conducted at a 746-bed academic medical centre, an alert threshold of **ESM $\geq$ 5** was selected using receiver operating characteristic (ROC) curve analysis [1]. At this threshold, the model demonstrated:

Metric	Value
Area Under ROC Curve (AUC)	0.834
Sensitivity	86.0%
Specificity	80.8%
Positive Predictive Value (PPV)	33.8%
Negative Predictive Value (NPV)	98.1%

These results indicate that the ESM is optimised as a screening tool, prioritising sensitivity and early detection rather than definitive diagnosis [1].

1.1 Functional Goals of the System

The primary functional goal of the Epic Sepsis Model is early identification of hospitalised patients at risk of developing sepsis, enabling clinicians to initiate timely evaluation and treatment. This goal is operationalised through continuous risk scoring and automated alerting within the EHR. More specifically, the system is designed to:

1. Detect elevated sepsis risk earlier than routine clinical recognition

By continuously analysing patient data, the ESM aims to identify deterioration before overt clinical signs prompt clinician action.

2. Prompt timely clinical intervention

When the ESM score exceeds the alert threshold, clinicians are notified and expected to assess the patient, consider sepsis protocols, and initiate appropriate management.

3. Reduce time to antibiotic administration

In the post-implementation phase of the validation study, the median time from alert to antibiotic administration decreased from 150 minutes to 90 minutes, and the proportion of patients receiving antibiotics within three hours increased from 55.6% to 69.8% [1].

4. Improve patient outcomes, particularly mortality

Among patients with ESM scores ≥ 5 who had not already received antibiotics, implementation of the alert system was associated with a 44% reduction in the odds of in-hospital sepsis related mortality (odds ratio 0.56; 95% CI 0.39–0.80) [1].

5. Integrate seamlessly into existing clinical workflows

The system is embedded within the Epic EHR and requires no separate interface, enabling widespread deployment with minimal disruption.

Ethical Relevance of the Functional Design

Although the ESM demonstrably improves early intervention and is associated with reduced mortality, its design raises ethical challenges that extend beyond functional performance. The system produces risk scores that significantly influence clinician behaviour, yet its internal reasoning is largely opaque. Clinicians remain legally and morally responsible for decisions, while the AI system shapes attention, urgency, and treatment timing. This creates potential risks of automation bias, over-reliance, and accountability gaps particularly in high-stakes clinical contexts [3, 5].

Chapter 2: Stakeholder Identification

The Epic Sepsis Model (ESM) operates within a complex hospital environment involving clinical personnel, patients, healthcare institutions, and technology providers. As a deployed AI-enabled CDSS, its ethical impact arises from interactions between these stakeholders rather than from the predictive model alone.

Primary Stakeholders

Primary stakeholders are those who directly interact with the system or are directly affected by its outputs.

1. Clinicians (Physicians and Nurses)

Category: Social, Technical, Legal

Clinicians are the primary users of the ESM. The system generates alerts when a patient's sepsis risk score exceeds a predefined threshold ($ESM \geq 5$), prompting clinical evaluation and intervention. Although clinicians retain final decision-making authority, the alerts influence attention, prioritisation, and timing of treatment. Clinicians are also legally accountable for patient outcomes, even when decisions are informed by AI-generated risk scores [3].

2. Patients

Category: Social, Ethical, Legal

Hospitalised patients are the subjects of the predictive model. They do not directly interact with the system but are affected by its outputs through changes in clinical behaviour. Patients bear the risks of *false positives* (unnecessary treatment, antibiotic overuse) and *false negatives* (missed or delayed diagnosis).

3. Hospital and Healthcare Providers

Category: Economic, Technical, Legal

Hospitals deploy the ESM as part of the Epic EHR infrastructure. They benefit from potential reductions in mortality, length of stay, and compliance with sepsis quality measures. At the same time, hospitals assume institutional liability and must manage alert fatigue, workflow disruption, and training requirements [6].

Secondary Stakeholders

4. Epic Systems Corporation

Category: Technical, Economic, Legal

Epic develops and maintains the proprietary ESM algorithm. Its design choices including feature selection, model updates, and alert presentation shape how clinicians interpret and act on risk scores.

5. Health Regulators and Accreditation Bodies

Category: Political, Legal

Regulatory bodies influence the adoption of sepsis detection systems through quality standards and patient safety regulations.

6. Society and Public Health Systems

Category: Social, Economic

At a population level, widespread deployment influences healthcare costs, antibiotic stewardship, and public trust in AI-assisted medicine.

7. Environmental and Infrastructure Considerations

Category: Environmental, Technical

Large-scale deployment across national hospital networks contributes to cumulative energy consumption and digital infrastructure expansion. Responsible digital health governance must consider long-term sustainability.

2.1 Ethical Goals of the System

The Epic Sepsis Model does not publish an explicit ethical framework. Public documentation and validation studies focus primarily on clinical effectiveness, workflow integration, and patient outcomes. There are no explicit public statements addressing transparency, accountability, or ethical safeguards beyond standard clinical governance practices. This absence is ethically relevant, as it requires ethical safeguards to be inferred from system behaviour and deployment structure rather than from explicitly articulated normative commitments [5].

2.2 Ethical Considerations

For this project, three schools of ethical thought are used to analyse the system: **Utilitarianism**, **Deontological Ethics**, and **Virtue Ethics**. These were selected because they capture outcome based, duty-based and professional character-based ethical concerns relevant to clinical AI systems.

1. Utilitarian Ethics (Outcome-Oriented)

Core Principle: Actions are ethically justified if they maximise overall benefit and minimise harm.

Relevant variables: Sepsis mortality rates, time to antibiotic administration, sensitivity and specificity of alerts, hospital-wide resource utilisation.

Ethical goals: Reduce sepsis-related mortality; enable earlier treatment for high-risk patients; improve population-level outcomes.

System behaviour: The ESM prioritises sensitivity (86.0%) over precision (PPV 33.8%) to ensure that most sepsis cases are detected early. This design accepts a higher false-positive rate to maximise overall survival benefit, consistent with utilitarian reasoning [1].

Ethical risk: Increased false positives may lead to unnecessary antibiotic use, increased costs, and alert fatigue, which may undermine long-term benefit.

2. Deontological Ethics (Duty and Responsibility)

Core Principle: Ethical action depends on adherence to duties, rules, and accountability, not outcomes alone.

Relevant variables: Alert thresholds, clinician accountability, opacity of model reasoning.

Ethical goals: Preserve clinician responsibility for decisions; avoid delegation of moral responsibility to the AI system; ensure clinicians can justify actions taken in response to alerts.

System behaviour: The ESM provides a numerical risk score but does not issue mandatory treatment recommendations. Clinicians are expected to evaluate alerts using clinical judgement.

Ethical risk: The system influences behaviour without providing transparent reasoning, creating a *responsibility gap* where clinicians remain accountable but lack full epistemic access to the basis of the alert [4, 5].

3. Virtue Ethics (Professional Character and Practice)

Core Principle: Ethical systems should support good professional judgement, not replace it.

Relevant variables: Human–AI interaction, alert frequency, workflow integration, training and education.

Ethical goals: Support prudent clinical decision-making; avoid deskilling and over-reliance on automated alerts; maintain clinician attentiveness and reflective judgement.

System behaviour: The ESM is integrated into routine workflows and requires clinicians to actively interpret and act on alerts.

Ethical risk: Frequent alerts and demonstrated reductions in time to antibiotics indicate strong behavioural influence, which may encourage automation bias over time [7].

Ethical Framework–Stakeholder Alignment

The three ethical schools correspond to different stakeholder perspectives. Utilitarian analysis primarily concerns patient outcomes and hospital-level mortality reduction. Deontological considerations are most salient for clinicians, who retain formal decision authority and legal responsibility for AI-informed actions. Virtue ethics focuses on the professional character of clinicians interacting repeatedly with alert systems, emphasising calibration, attentiveness, and avoidance of automation bias.

2.3 Values

For **clinicians**, the primary value is patient welfare, pursued through informed and independent professional judgement. This aligns with virtue ethics and deontological accountability requirements, but is threatened by automation bias and the responsibility gap. For **patients**, the central value is safety and equitable access to timely care, aligning with both utilitarian outcomes and the justice principle implicit in deontological ethics. For **hospitals**, values include operational efficiency, liability management, and regulatory compliance, which may at times conflict with full transparency regarding model limitations. For **Epic Systems**, commercial and technical values—model confidentiality, proprietary architecture—create structural tension with the transparency values of clinicians and patients.

The most significant negotiation required is between Epic's proprietary opacity and the clinicians' epistemic need for accountability. Patients' safety values ought to be prioritised, as they bear the direct risk of both false positive and false negative outcomes.

Chapter 3: Implementation

This chapter contains a link to the repository, where the code is kept.

Link to Repository: <https://github.com/AshwinPrasanth/AI-Ethics>

3.1 Description of Approach

Model Abstraction

The Epic Sepsis Model is a proprietary, data-driven predictive system. This implementation does not attempt model reproduction. Instead, it preserves the operational invariants that define the system's decision structure:

- Continuous numerical risk scoring
- Threshold-based alert triggering (Score ≥ 5)
- Screening-oriented behaviour (high sensitivity design)
- Non-zero false positive possibility
- Human-in-the-loop decision authority
- Integration into clinician workflow

The system therefore represents a structural abstraction of a sub-symbolic CDSS, sufficient for ethical validation experiments.

Domain Representation

Each synthetic inpatient instance is represented as:

$x = (T, HR, Lactate, WBC, SBP, PriorAntibiotics)$

where T = Temperature, HR = Heart rate, $Lactate$ = Serum lactate, WBC = White blood cell count, SBP = Systolic blood pressure. The risk score is computed as $Score = \sum_i w_i x_i$ with fixed weights. The purpose is not predictive realism, but simulation of sub-symbolic weighted aggregation of heterogeneous physiological signals.

Functional Goal Encoding

Functional goals are encoded in `functional_validation.py`:

- **F1: Continuous Risk Stratification** — The system produces a real-valued risk score for each patient. High-risk synthetic cases produce $\text{Score} \geq 5$; moderate-risk cases produce $\text{Score} < 5$.
- **F2: Deterministic Alert Trigger** — $\text{Alert} = (\text{Score} \geq 5)$. The threshold of 5 aligns with the published ESM validation cut-point.
- **F3: Workflow Activation** — If $\text{Alert} = \text{True}$, the system simulates escalation to clinician and a decision node requiring response.

Ethical Goal Formalization (Simulation Layer)

Ethical goals are encoded as executable constraint checks in `ethical_validation.py`:

Code	Ethical Goal	Condition
E1	False Positive Monitoring	$\text{Alert} = \text{True} \wedge \text{GroundTruth} = \text{False}$
E2	Over-Reliance Detection	$\text{Alert} = \text{True} \wedge \text{Accept} \wedge \text{GroundTruth} = \text{False}$
E3	Under-Reliance Detection	$\text{Alert} = \text{True} \wedge \text{ClinicianAction} = \text{Ignore}$
E4	Responsibility Gap	$\text{Alert} = \text{True} \wedge \text{ExplanationUnavailable}$
E5	Alert Fatigue Monitoring	$\text{Alert counter} > 5$
E6	Governance Constraint	$\text{Alert} \rightarrow \text{Acknowledged}$
E7	Data Quality Failure	Corrupted/incomplete inputs
E8	Age-Based Fairness	$\text{Equivalent state} \rightarrow \text{equal alerts}$

Note on E4 (Responsibility Gap): In the implementation, `generate_explanation()` always returns a feature contribution dictionary, meaning `ExplanationUnavailable` is always `False`. This is a deliberate abstraction boundary: in real ESM deployment the gap is structurally present. The Z3 layer addresses this formally.

Formal Logical Verification (Z3 SMT Layer)

Beyond simulation, selected ethical constraints are encoded as logical formulas and analysed using the Z3 SMT solver in `z3_ethics_verification.py`. Constraints are registered using `assert_and_track`, enabling UNSAT core extraction.

- **SAT Case: Duty-of-Care Consistency** — $(\text{Alert} \wedge \text{GroundTruth}) \rightarrow \text{ClinicianResponded}$. Z3 returns SAT, demonstrating logical consistency.
- **UNSAT Case: Capacity Conflict** — $\text{Treat}_A \wedge \text{Treat}_B$ under $\neg(\text{Treat}_A \wedge \text{Treat}_B)$. Unsats Core: `{Ethical_Treat_A, Ethical_Treat_B, Capacity_Constraint}`.
- **UNSAT Case: Age-Based Fairness Violation** — $\text{EquivalentClinicalState} \rightarrow (\text{Alert}_{\text{elderly}} = \text{Alert}_{\text{young}})$. Unsats Core: `{Elderly_Alert_True, Equivalent_Group_True, Fairness_Symmetry, Young_Alert_False}`.
- **UNSAT Case: Governance Violation** — $\text{Alert} \rightarrow \text{Acknowledged}$. Unsats Core: `{Governance_Rule, Alert_Occurred, No_Acknowledgement}`.

3.2 Points of Conflict

Conflict 1: Fairness vs. Clinical Equivalence

Original design decision: The initial scoring model treated all patients identically, using only physiological variables with no subgroup differentiation.

Value conflict identified: Deontological and virtue ethics considerations raised the question of whether a system without explicit fairness testing could inadvertently produce discriminatory outcomes for elderly patients with atypical sepsis physiology. The absence of fairness testing is itself an ethical risk.

Design change: Two additional synthetic patient scenarios were introduced `elderly_high_risk_patient()` and `young_high_risk_patient()`—with identical clinical values but different age group labels. A fairness check (E8) and a formal Z3 fairness symmetry constraint were added.

Stakeholders who benefit: Patients (particularly elderly patients); regulators and accreditation bodies.

Conflict 2: Transparency vs. Proprietary Opacity

Original design decision: The initial abstraction included a `generate_explanation()` function treating interpretability as an available capability.

Value conflict identified: This misrepresented the real ESM, which does not expose its internal reasoning to clinicians. The deontological concern around accountability required the abstraction to reflect this opacity [5].

Design change: The E4 responsibility gap constraint was retained but documented as a known abstraction limitation. The Z3 layer was extended to formally model governance and accountability constraints.

Stakeholders who benefit: Clinicians (explicit acknowledgement that accountability gaps exist); patients (safety depends on clinicians understanding AI limits).

Conflict 3: Sensitivity Optimisation vs. Alert Fatigue

Original design decision: The alert threshold was fixed at $\text{Score} \geq 5$ without analysis of threshold variation effects.

Value conflict identified: Optimising for sensitivity necessarily increases false positives, creating a direct conflict between population-level mortality benefit and individual clinician workflow sustainability.

Design change: A threshold trade-off simulation was added (`simulate_threshold_tradeoff()`), evaluating alert behaviour at thresholds of 4, 5, and 7. An alert fatigue counter (E5) was added to accumulate alert frequency and flag when the threshold is exceeded.

Stakeholders who benefit: Clinicians (alert burden made visible); hospitals (threshold as governance lever); patients (indirectly, through attentive clinicians).

3.3 Evidence

Functional Validation Output

Execution of `python functional_validation.py`:

```
1 --- Functional Validation ---
2 High Risk Score: 5.750000000000001
3 Alert Triggered: True
4 Moderate Risk Score: 2.9800000000000013
5 Alert Triggered: False
```

A clinically high-risk patient profile produces a score of 5.75, correctly triggering an alert. A moderate-risk profile scores 2.98, correctly suppressing the alert. This validates **F1** (continuous risk stratification) and **F2** (deterministic alert trigger).

A terminal window screenshot showing the execution of the functional validation script. The prompt is '@AshwinPrasanth → /workspaces/AI--Ethics (main) \$'. The command is '/usr/bin/python3 /workspaces/AI--Ethics/epic-ethical-validation/functional_validation.py'. The output is: '--- Functional Validation ---', 'High Risk Score: 5.750000000000001', 'Alert Triggered: True', 'Moderate Risk Score: 2.9800000000000013', and 'Alert Triggered: False'.

```
@AshwinPrasanth → /workspaces/AI--Ethics (main) $ /usr/bin/python3 /workspaces/AI--Ethics/epic-ethical-validation/functional_validation.py
--- Functional Validation ---
High Risk Score: 5.750000000000001
Alert Triggered: True
Moderate Risk Score: 2.9800000000000013
Alert Triggered: False
```

Figure 3.1: Functional validation terminal output

Ethical Validation Output

Execution of `python ethical_validation.py`:

```
1 --- Over-reliance Scenario ---
2 Over-reliance detected: True
3
4 --- Basic False Positive Case ---
5 False Positive Rate: 0.0
6
7 --- Aggressive False Positive Case ---
8 Aggressive False Positive Score: 5.402000000000001
9 Alert Triggered: True
10 False Positive Rate: 1.0
11
12 --- Under-reliance Scenario ---
13 Alert: True
14 Clinician responded: False
15 Ethical goal satisfied: False
16
17 --- Responsibility Gap Test ---
18 Alert: True
19 Explanation available: True
20 Responsibility gap detected: False
21
22 --- Threshold Trade-Off Simulation ---
23 Threshold: 4 | Alert: True | FPR: 1.0
24 Threshold: 5 | Alert: True | FPR: 1.0
25 Threshold: 7 | Alert: False | FPR: 0.0
26
```

```

27 --- Alert Fatigue Simulation ---
28 Alert fatigue triggered: True
29
30 --- Governance Check ---
31 Governance constraint satisfied: False
32
33 --- Data Quality Failure Scenario ---
34 Data quality failure detected: Missing required feature: temperature
35
36 --- Fairness Across Subgroups ---
37 Elderly Alert: True
38 Young Alert: True
39 No subgroup disparity detected.

```

```

@AshwinPrasanth → /workspaces/AI--Ethics (main) $ /usr/bin/python3 /workspaces/AI--Ethics/epic-ethical-validation/ethical_validation.py

--- Over-reliance Scenario ---
Over-reliance detected: True

--- Basic False Positive Case ---
False Positive Rate: 0.0

--- Aggressive False Positive Case ---
Aggressive False Positive Score: 5.4020000000000001
Alert Triggered: True
False Positive Rate: 1.0

--- Under-reliance Scenario ---
Under-reliance scenario:
Alert: True
Clinician responded: False
Ethical goal satisfied: False

--- Responsibility Gap Test ---
Alert: True
Explanation available: True
Responsibility gap detected: False

--- Threshold Trade-Off Simulation ---
Threshold: 4
Alert Triggered: True
False Positive Rate: 1.0

Threshold: 5
Alert Triggered: True
False Positive Rate: 1.0

Threshold: 7
Alert Triggered: False
False Positive Rate: 0.0

--- Reliance Calibration Test ---
Reliance type: Blind Compliance

--- Alert Fatigue Simulation ---
Alert fatigue triggered: True

--- Governance Check ---
Governance constraint satisfied: False

--- Data Quality Failure Scenario ---
Data quality failure detected: Missing required feature: temperature

--- Fairness Across Subgroups ---
Elderly Alert: True
Young Alert: True
No subgroup disparity detected.

```

Figure 3.2: Ethical validation terminal output

Z3 Formal Verification Output

Execution of `python z3_ethics_verification.py`:

```

1 --- SAT: Duty of Care Scenario ---
2 Solver result: sat
3 Model: [Alert = True, ClinicianResponded = True,
4         Duty_of_Care = True, GroundTruth = True]

```

```

5
6 --- UNSAT: Capacity Conflict ---
7 Solver result: unsat
8 Unsat Core: [Ethical_Treat_A, Ethical_Treat_B, Capacity_Constraint]
9
10 --- UNSAT: Fairness Violation ---
11 Solver result: unsat
12 Unsat Core: [Elderly_Alert_True, Equivalent_Group_True,
13             Fairness_Symmetry, Young_Alert_False]
14
15 --- UNSAT: Governance Violation ---
16 Solver result: unsat
17 Unsat Core: [Governance_Rule, No_Acknowledgement, Alert_Occurred]

```

```

@AshwinPrasanth → /workspaces/AI--Ethics/epic-ethical-validation (main) $ python3 z3_ethics_verification.py

--- SAT: Duty of Care Scenario ---
Solver result: sat
Model: [Alert = True,
ClinicianResponded = True,
Duty_of_Care = True,
GroundTruth = True]

--- UNSAT: Capacity Conflict ---
Solver result: unsat
Unsat Core: [Ethical_Treat_A, Ethical_Treat_B, Capacity_Constraint]

--- UNSAT: Fairness Violation ---
Solver result: unsat
Unsat Core: [Elderly_Alert_True,
Equivalent_Group_True,
Fairness_Symmetry,
Young_Alert_False]

--- UNSAT: Governance Violation ---
Solver result: unsat
Unsat Core: [Governance_Rule, No_Acknowledgement, Alert_Occurred]

```

Figure 3.3: Z3 formal verification output

Audit Log

```

1 2026-02-28 17:46:42 - WARNING - Over-reliance scenario detected.
2 2026-02-28 17:46:42 - INFO - False Positive Test - Score: 5.402, Alert:
  True, FPR: 1.0
3 2026-02-28 17:46:42 - WARNING - Under-reliance detected - Alert: True,
  Responded: False
4 2026-03-02 10:23:27 - WARNING - Over-reliance scenario detected.
5 2026-03-02 10:23:27 - ERROR - Data quality failure: Missing required
  feature: temperature
6 2026-03-02 10:26:22 - WARNING - Under-reliance detected - Alert: True,
  Responded: False

```

```

audit.log
1 2026-02-11 20:08:28,079 - WARNING - Over-reliance scenario detected.
2 2026-02-11 20:08:28,079 - INFO - False Positive Test - Score: 5.402000000000001, Alert: True, FPR: 1.0
3 2026-02-11 20:08:28,080 - WARNING - Under-reliance detected - Alert: True, Clinician responded: False
4 2026-03-09 10:52:32,101 - WARNING - Over-reliance scenario detected.
5 2026-03-09 10:52:32,111 - INFO - False Positive Test - Score: 5.402000000000001, Alert: True, FPR: 1.0
6 2026-03-09 10:52:32,111 - WARNING - Under-reliance detected - Alert: True, Clinician responded: False
7 2026-03-09 10:52:32,111 - ERROR - Data quality failure detected: Missing required feature: temperature
8 2026-03-09 18:59:45,694 - WARNING - Over-reliance scenario detected.
9 2026-03-09 18:59:45,694 - INFO - False Positive Test - Score: 5.402000000000001, Alert: True, FPR: 1.0
10 2026-03-09 18:59:45,695 - WARNING - Under-reliance detected - Alert: True, Clinician responded: False
11 2026-03-09 18:59:45,695 - ERROR - Data quality failure detected: Missing required feature: temperature
12

```

Figure 3.4: Audit log showing timestamped ethical violation trace

Chapter 4: Claude Says. . .

Model used: Claude Sonnet (claude.ai, free tier)

Date of debate: March 2026

Note: All responses were elicited using the free-tier web interface at claude.ai. No system prompt modifications were applied. Each argument was presented as a standalone prompt in a fresh conversation to avoid contextual carry-over between arguments.

Overview

This chapter records a debate with Claude across the four ethical conflict points identified in Chapter 3. Each argument was prompted separately, asking for at least two distinct positions with reasoning. Claude's responses are reproduced in full and then examined critically. Where the reasoning is weak, incomplete, or logically flawed, this is noted directly.

Argument 1: Sensitivity Optimisation vs Alert Fatigue — Should the Threshold Be Raised?

Prompt

"In a deployed AI sepsis detection system, the alert threshold is set at Score ≥ 5 , optimising for high sensitivity (86%) but producing a positive predictive value of only 33.8%. This means roughly two thirds of alerts are false positives. From an ethical standpoint, should the threshold be raised to reduce alert fatigue, even if some sepsis cases are missed? Give me at least two distinct actions or non-actions you would recommend, and explain the reasoning behind each. Please keep your response to 300–400 words total."

Prompt: In a deployed AI sepsis detection system, the alert threshold is set at $\text{Score} \geq 5$, optimising for high sensitivity (86%) but producing a positive predictive value of only 33.8%. This means roughly two thirds of alerts are false positives. From an ethical standpoint, should the threshold be raised to reduce alert fatigue, even if some sepsis cases are missed? Give me at least two distinct actions or non-actions you would recommend, and explain the reasoning behind each. Please keep your response to 300–400 words total.

Claude's Response — Ethical Analysis: Sepsis Alert Threshold

Recommendation 1: Do not raise the threshold unilaterally — instead, stratify alerts by context. Rather than a blunt threshold increase, the system should implement tiered alerting based on clinical setting. An ICU team managing critically ill patients can act on lower-confidence alerts; a general ward with stretched staffing cannot. Deploying a

higher threshold (say, ≥ 7) only in lower-acuity settings preserves sensitivity where it matters most while reducing noise where alert fatigue is most damaging. The ethical justification is proportionality.

Recommendation 2: Measure and act on alert fatigue empirically before changing the threshold at all. Before adjusting the threshold, the deploying organisation should audit clinician override rates, response times, and documented sepsis cases acted upon versus ignored. If clinicians are already dismissing 70% of alerts reflexively, the current threshold may already be functionally equivalent to a higher one — just with hidden harm. The ethical principle here is epistemic humility.

Underlying principle: Optimising a threshold is not purely a statistical exercise — it is a values-laden decision about whose risk is acceptable and who bears the burden of error. That decision should involve clinicians, patients, and ethicists, not only the AI development team.

Figure 4.1: Claude's response to Argument 1 (claude.ai, free tier, March 2026)

Claude's Response

Recommendation 1: Do not raise the threshold unilaterally — instead, stratify alerts by context.

Rather than a blunt threshold increase, the system should implement tiered alerting based on clinical setting. An ICU team managing critically ill patients can act on lower-confidence alerts; a general ward with stretched staffing cannot. A uniform $\text{Score} \geq 5$ rule ignores this reality. Deploying a higher threshold (say, ≥ 7) only in lower-acuity settings preserves sensitivity where it matters most while reducing noise where alert fatigue is most damaging. The ethical justification is proportionality.

Recommendation 2: Measure and act on alert fatigue empirically before changing the threshold at all.

Before adjusting the threshold, the deploying organisation should audit clinician override rates, response times, and documented sepsis cases acted upon versus ignored. If clinicians are already dismissing 70% of alerts reflexively, the current threshold may already be functionally equivalent to a higher one. Raising the threshold without this data risks enshrining a guess as policy. The ethical principle here is epistemic humility.

Underlying principle: Optimising a threshold is not purely a statistical exercise—it is a values-laden decision about whose risk is acceptable. That decision should involve clinicians, patients, and ethicists, not only the AI development team.

Analysis

Recommendation 2 is the stronger of the two. Grounding threshold decisions in local audit data before making any change is reasonable, and the point about clinicians already dismissing most alerts reflexively is a real phenomenon.

Recommendation 1 is where the reasoning breaks down. Claude suggests raising the threshold on general wards, framing this as proportionality. The problem is that a patient on a general ward with a score of 6 has the same physiological state as an ICU patient with a score of 6. Under this proposal, only the ICU patient receives an alert. General ward patients also tend to have less continuous monitoring than ICU patients, making a missed deteriorating case there arguably more dangerous. The logic runs in the opposite direction to what Claude claims.

Logical error: Recommendation 1 treats ward type as a reliable proxy for the consequences of false positives and false negatives. It is not, and the reasoning produces exactly the inequity it claims to avoid.

Argument 2: Transparency vs Proprietary Opacity — Is the Responsibility Gap Defensible?

Prompt

“The Epic Sepsis Model generates a risk score but does not expose its internal reasoning to clinicians. Clinicians remain legally and morally accountable for decisions made in response to AI alerts, yet cannot fully explain the basis of those alerts. Is this ethically acceptable? Give me at least two distinct positions with the reasoning behind each. Please keep your response to 300–400 words total.”

Prompt: The Epic Sepsis Model generates a risk score but does not expose its internal reasoning to clinicians. Clinicians remain legally and morally accountable for decisions made in response to AI alerts, yet cannot fully explain the basis of those alerts. Is this ethically acceptable? Give me at least two distinct positions — one defending the current opaque design and one arguing for mandatory explainability — with the reasoning behind each.

Please keep your response to 300–400 words total.

Claude’s Response — Explainability in the Epic Sepsis Model: Two Positions

Position 1: Opacity is defensible — clinicians are not obligated to understand the tool’s internals. A clinician using an opaque sepsis alert is in a structurally similar position to one interpreting a lab result: they do not need to understand the chemistry of a troponin assay to act appropriately on an elevated reading. If clinicians know the score has 86% sensitivity and a 33.8% PPV, they have actionable calibration information. Demanding full mechanistic transparency may be counterproductive — complex feature interactions can produce explanations that are technically accurate but cognitively misleading.

Position 2: Opacity is ethically unacceptable when accountability remains with the clinician. Clinicians bear full legal and moral responsibility for outcomes, yet are denied access to the reasoning that generated the alert. This is not analogous to a lab test — a troponin value is a direct measurement of a biological quantity, whereas an AI score is an inference built on weighted feature combinations that may encode biases or training artefacts invisible to the end user. Meaningful accountability requires meaningful access.

Unresolved tension: Both positions share a concern for patient safety but disagree on whether opacity or false explainability poses the greater risk. That disagreement demands empirical study of how clinicians actually use explanations when provided.

Figure 4.2: Claude’s response to Argument 2 (claude.ai, free tier, March 2026)

Claude’s Response

Position 1: Opacity is defensible — clinicians are not obligated to understand the tool’s internals.

A clinician using an opaque sepsis alert is in a structurally similar position to one interpreting a lab result: they do not need to understand the chemistry of a troponin assay to act appropriately on an elevated reading. What matters is that the tool’s population-level performance is validated, communicated, and understood. Demanding full mechanistic transparency may be counterproductive—complex feature interactions in ML models can produce explanations that are technically accurate but cognitively misleading.

Position 2: Opacity is ethically unacceptable when accountability remains with the clinician.

The asymmetry is genuinely troubling. Clinicians bear full legal and moral responsibility for outcomes, yet are denied access to the reasoning that generated the alert they must respond to. This is not analogous to a lab test a troponin value is a direct measurement of a biological quantity, whereas an AI score is an inference built on weighted feature combinations that may encode biases or training artefacts invisible to the end user. Meaningful accountability requires meaningful access [5].

Unresolved tension: Both positions share a concern for patient safety but disagree on whether opacity or false explainability poses the greater risk.

Analysis

Position 2 correctly identifies the core problem. Position 1 fails via a false analogy. A troponin result is a direct measurement of a biological quantity with known, traceable error sources reagent failure, sample contamination, assay interference. An ESM score is a learned inference constructed from weighted combinations of roughly 80 variables, trained on historical data that may encode past care inequities. When the ESM is systematically wrong for a particular patient group, that error is invisible to the clinician at the bedside.

Logical error: False analogy. A direct biological measurement and a black-box ML inference have categorically different error structures and accountability implications.

Argument 3: Fairness Across Age Subgroups — Should Testing Be Mandatory Pre-Deployment?**Prompt**

“A sepsis risk model is trained on a large patient population but does not explicitly test for performance differences across age groups. Elderly patients often present with atypical sepsis physiology and may be systematically underscored. Should fairness testing across age subgroups be a mandatory requirement before deploying such a system? Give me at least two distinct actions or non-actions. Please keep your response to 300–400 words total.”

Prompt: A sepsis risk model is trained on a large patient population but does not explicitly test for performance differences across age groups. Elderly patients often present with atypical sepsis physiology and may be systematically underscored. Should fairness testing across age subgroups be a mandatory requirement before deploying such a system? Give me at least two distinct actions or non-actions, with reasoning for each. Please keep your response to 300–400 words total.

Claude’s Response — Fairness Testing Across Age Subgroups: Two Positions

Action 1: Mandate subgroup fairness testing as a pre-deployment requirement. Deploying a model without testing performance across age groups is not a neutral omission — it is a decision to accept unknown differential harm. Elderly patients present particular physiological challenges: blunted fever responses, atypical inflammatory markers, and baseline comorbidities that may systematically suppress risk scores. The ethical principle of justice requires that foreseeable disparate impacts be identified before deployment, not discovered retrospectively through adverse outcomes.

Non-action 2: Do not delay deployment pending perfect subgroup parity — but require post-deployment monitoring. Mandatory pre-deployment fairness testing risks becoming a barrier that keeps demonstrably useful tools out of clinical practice. A model with imperfect subgroup performance may still reduce net sepsis mortality compared to no decision support at all. The more proportionate requirement is conditional deployment alongside a mandatory post-deployment surveillance protocol stratified by age.

Shared commitment: Both recommendations reject deploying without any fairness consideration. What is ethically indefensible is treating subgroup performance as an afterthought — something to investigate only after a pattern of missed diagnoses forces the question.

Figure 4.3: Claude’s response to Argument 3 (claude.ai, free tier, March 2026)

Claude’s Response

Action 1: Mandate subgroup fairness testing as a pre-deployment requirement.

Deploying a model without testing performance across age groups is not a neutral omission—it is a decision to accept unknown differential harm. Elderly patients present particular physiological challenges: blunted fever responses, atypical inflammatory markers, and baseline comorbidities that may systematically suppress risk scores. The justice principle requires that foreseeable disparate impacts be identified before deployment.

Non-action 2: Do not delay deployment pending perfect subgroup parity — require post-deployment monitoring instead.

Mandatory pre-deployment fairness testing risks becoming a barrier that keeps useful tools out of clinical practice. A model with imperfect subgroup performance may still reduce net sepsis mortality compared to no decision support at all. The more proportionate requirement is conditional deployment alongside a mandatory post-deployment surveillance protocol stratified by age.

Analysis

Action 1 is well reasoned. Non-action 2 has two critical problems. First, the comparison is wrong: the counterfactual is not zero clinical decision support—most hospitals already operate structured sepsis screening protocols. Second, post-deployment monitoring means evidence of differential harm is collected from patients who are already being missed. Those are not data points in a quality improvement exercise; they are people who did not receive timely alerts.

Logical error: The “better than nothing” framing rests on a false counterfactual and treats foreseeable harm to elderly patients as an administrative cost rather than an ethical failure.

Argument 4: Automation Bias — Design Problem, Training Problem, or Inherent Risk?

Prompt

“Studies show that when clinicians are repeatedly exposed to AI alerts, they tend to follow them without independent assessment—automation bias. In a high-stakes clinical setting like sepsis detection, is automation bias primarily a training problem, a system design problem, or an inherent risk? Give me at least two distinct recommended actions or non-actions. Please keep your response to 300–400 words total.”

Prompt: Studies show that when clinicians are repeatedly exposed to AI alerts, they tend to follow them without independent assessment — a phenomenon called automation bias. In a high-stakes clinical setting like sepsis detection, is automation bias primarily a training problem, a system design problem, or an inherent risk of deploying AI in clinical workflows at all? Give me at least two distinct recommended actions or non-actions, with the reasoning behind each. Please keep your response to 300–400 words total.

Claude’s Response — Automation Bias in Sepsis AI: Diagnosis and Response

Position 1: Automation bias is primarily a system design problem — and design can mitigate it. A binary alarm invites passive compliance; a display that shows contributing factors and prompts documentation of independent clinical reasoning invites engagement. Interface friction — requiring a clinician to actively confirm or override with a brief rationale — reduces reflexive compliance without meaningfully slowing response times. If automation bias is reliably produced by particular interface patterns, deploying those patterns unreflectively is an institutional choice, not an inevitability.

Action 2: Implement structured training, but treat it as necessary rather than sufficient. Training clinicians to recognise automation bias and practice deliberate override decisions is ethically mandatory. However, cognitive shortcuts under time pressure and fatigue are predictable features of human performance in high-load environments, not failures of knowledge. Training addresses the skill gap; it does not address the conditions that make bias adaptive in the first place.

Unresolved question: Automation bias may be partially inherent to any system where AI output is faster and less effortful than independent assessment. The honest institutional response is to explicitly accept residual bias as a known risk and maintain human staffing levels sufficient to make genuine clinical reasoning possible.

Figure 4.4: Claude’s response to Argument 4 (claude.ai, free tier, March 2026)

Claude’s Response

Position 1: Automation bias is primarily a system design problem — and design can mitigate it.

A binary alarm invites passive compliance; a display showing contributing factors and prompting documentation of independent clinical reasoning invites engagement. Interface friction—requiring a clinician to actively confirm or override with a brief rationale—reduces reflexive compliance without meaningfully slowing response times.

Action 2: Implement structured training, but treat it as necessary rather than sufficient.

Training clinicians to recognise automation bias, understand the model’s known limitations, and practice deliberate override decisions is ethically mandatory. However, training alone is insufficient. Cognitive shortcuts under time pressure and fatigue are predictable features of human performance in high-load environments, not failures of knowledge [7].

Unresolved question: Automation bias may be partially inherent to any system where AI output is faster and less effortful than independent assessment. The honest institutional response is to accept residual bias as a known risk and maintain staffing levels sufficient to make genuine clinical reasoning possible.

Analysis

The two positions complement each other: design and training address different failure modes. However, having raised the possibility of inherent automation bias, Claude does not reason through it. More substantially, Claude does not engage with the virtue ethics dimension at all. The deeper concern is what repeated reliance on AI alerts does to clinical judgement over time—a clinician who has spent years following ESM alerts without practising independent assessment may have lost the diagnostic instinct that makes meaningful override possible [7].

Logical error: The response names the hardest problem and defers resolution. Omission of virtue ethics is notable given it was one of the three frameworks selected for this project.

Summary of Identified Errors

Argument	Claude’s Recommendation Error / Weakness	
1 — Threshold	Stratify by ward type	Inverts actual risk distribution
2 — Opacity	Lab test analogy	False analogy; categorically different error structures
3 — Fairness	Post-deployment monitoring	False counterfactual; treats harm as administrative
4 — Automation bias	Design plus training	Names the hardest problem then defers; omits virtue

Chapter 5: Teammate #1 Says...

This chapter is written solely by team member #1.

Selected Argument

This chapter argues against Claude's Position 1 in Argument 2, which defended the opacity of the Epic Sepsis Model by claiming that clinicians interpreting an AI risk score are in a structurally similar position to clinicians interpreting a laboratory result. This analogy is false, and accepting it conceals a specific accountability failure that affects every alert the system generates.

The Analogy Does Not Hold

Claude's argument rests on the claim that a clinician does not need to understand the internal mechanics of a diagnostic tool to act appropriately on its output. The distinction that matters is not whether the clinician understands the mechanics. It is whether the source of error is *traceable* when the tool is wrong. A troponin result is a direct measurement of a biological quantity with documented, investigable failure modes: reagent degradation, sample haemolysis, assay interference. When a troponin is anomalous, the clinician and the laboratory can reason about why.

The ESM score is a learned inference from a proprietary model combining roughly 80 variables through undisclosed statistical relationships. Those relationships may encode documentation biases, historical care inequities, or population-specific confounders. When the ESM is systematically wrong for a particular patient group, that error is invisible at the point of care. There is no equivalent of checking the reagent batch. The clinician receives a number with no accessible basis for disagreement [5].

What the Analogy Conceals

The practical consequence is that clinicians are left accountable for decisions they cannot fully reason about. Consider this scenario: a clinician receives an alert for an elderly patient with atypical presentation, assesses the patient independently, decides the score does not reflect the clinical picture, and does not escalate. The patient later deteriorates.

The clinician's defence depends on articulating not only what they observed, but why they disagreed with the score. Under an opaque system, the second part of that defence has no grounding. They cannot say what the model weighted, what features drove the score, or whether the model had ever been validated on patients with this presentation.

The Structural Nature of the Gap

The responsibility gap is not an occasional scenario. It is the default operating state of the system. The E4 constraint in the ethical validation layer captures this directly:

```
1 def responsibility_gap(alert, explanation_available):  
2     if alert and not explanation_available:  
3         return True  
4     return False
```

In the real ESM deployment, `explanation_available` is `False` for every alert the system fires. The function therefore returns `True` on every single alert. This is a structural property of how the system is designed and deployed.

Claude argues that population-level metrics such as 86% sensitivity and 33.8% PPV provide sufficient calibration. Population-level statistics say nothing about the basis for any individual alert. A clinician told the model has 86% sensitivity knows that roughly one in seven sepsis cases will be missed overall. They do not know whether *this patient* is in that one in seven, or why. That gap is not closed by statistical disclosure [4].

An Internal Contradiction

Position 1 also contradicts what Claude acknowledged in Argument 4. The defence of opacity depends on clinicians consciously using population-level metrics to calibrate their response to each alert. But in Argument 4, Claude accepted that clinicians under time pressure and fatigue default to following alerts without independent assessment. These two pictures of clinical behaviour cannot both be true. Claude cannot use the first picture to justify opacity while accepting the second as the operational reality.

Conclusion

The lab test analogy fails because it treats traceability of error as irrelevant to accountability. It is not. What the ESM creates is uncertainty whose source is proprietary, undisclosed, and structurally inaccessible. Accepting Claude's position means accepting that as a sufficient basis for full clinical accountability. It is not [3, 5].

Chapter 6: Teammate #2 Says...

This chapter is written solely by team member #2.

Selected Argument

This chapter argues against Claude's Non-action 2 in Argument 3, which proposed that mandatory pre-deployment fairness testing should be replaced by conditional deployment with post-deployment monitoring. This position is ethically indefensible, and the formal verification implemented in this project demonstrates why.

The Counterfactual Is Wrong

The comparison Claude makes is between a fairness-untested model and no clinical decision support whatsoever. This is not the real choice. Most hospital systems deploying the ESM already operate structured sepsis screening protocols, early warning scores, and nursing assessment frameworks. The counterfactual is not an empty clinical environment—it is an environment with existing, if imperfect, decision support that does not carry the same untested subgroup risk.

The utilitarian calculation Claude implies only holds if the baseline is zero. It does not hold against an existing standard of care [6].

Post-Deployment Monitoring Is Not a Safeguard

A post-deployment surveillance protocol collects outcome data from patients who are already being managed under the untested system. If the model systematically underscores elderly patients with atypical sepsis presentation, those patients receive delayed or missed alerts while the institution accumulates evidence to confirm the disparity. The monitoring process does not prevent that harm. It measures it retrospectively.

Claude's framing uses the language of surveillance and tolerances, giving the impression of rigorous governance. What it describes is accepting foreseeable differential harm to a specific patient population as the price of early deployment, and then documenting the consequences.

The Violation Is Detectable Before Deployment

The strongest objection is that the epistemic justification for post-deployment monitoring does not hold. Claude implies the fairness violation cannot be known in advance. The Z3 verification layer in this project shows otherwise. The fairness constraint is encoded as a logical formula and

verified formally before any patient data is involved:

```
1 def unsat_fairness_violation():
2     s = z3.Solver()
3     Equivalent_Group = z3.Bool("Equivalent_Group")
4     elderly_alert = z3.Bool("elderly_alert")
5     young_alert = z3.Bool("young_alert")
6
7     s.assert_and_track(
8         z3.Implies(Equivalent_Group, elderly_alert == young_alert),
9         "Fairness_Symmetry"
10    )
11    s.assert_and_track(Equivalent_Group, "Equivalent_Group_True")
12    s.assert_and_track(elderly_alert, "Elderly_Alert_True")
13    s.assert_and_track(z3.Not(young_alert), "Young_Alert_False")
14
15    result = s.check()
16    if result == z3.unsat:
17        print("Unsat Core:", s.unsat_core())
```

This returns UNSAT with the core:

```
1 [Elderly_Alert_True, Equivalent_Group_True, Fairness_Symmetry,
   Young_Alert_False]
```

The violation scenario two clinically equivalent patients receiving different alert outcomes is formally proven to be logically inconsistent with the fairness constraint *before a single real patient is involved*. The precondition for Claude's argument that the violation cannot be known in advance is false. Pre-deployment formal verification closes exactly the epistemic gap that Claude uses to justify deferring the test [3].

Conclusion

Conditional deployment does not balance pragmatism against principle. It defers a test that can be conducted before deployment and accepts the harm that accrues in the interim as an administrative inevitability. The justice principle requires that foreseeable disparate impacts be identified before they occur, not after. The implementation in this project demonstrates that the necessary verification is technically feasible. Claude's Non-action 2 has no remaining justification.

Chapter 7: Conclusions

This project examined the Epic Sepsis Model through three ethical frameworks: utilitarianism, deontological ethics, and virtue ethics. Each framework illuminated a different dimension of the system's ethical profile. Utilitarian analysis surfaced the tension between sensitivity optimisation and false positive burden. Virtue ethics raised the longer-term concern about automation bias and clinical deskilling. Deontological ethics identified the accountability and governance failures that sit at the core of the system's design.

Of the three, deontological ethics proved the most tractable for both expressing the problems and implementing solutions. The reason is structural. Deontological reasoning operates through obligations and rules: if an alert fires, it must be acknowledged; if a clinician is accountable, they must have epistemic access; if two patients are clinically equivalent, they must receive equivalent treatment. These translate directly into executable constraints. Every ethical goal in the implementation—from the responsibility gap check to the governance constraint to the fairness symmetry rule—is a formalised duty of this kind. The Z3 verification layer makes this particularly clear: encoding an ethical obligation as a logical formula and checking whether it is satisfiable is deontological reasoning in a formal system [3].

Utilitarianism was useful for framing population-level trade-offs, particularly around threshold configuration and false positive rates, but harder to implement rigorously. Meaningful utilitarian validation would require outcome data across real patient populations, which a simulation abstraction cannot provide. The threshold trade-off simulation approximates this but cannot quantify net mortality benefit, which limits how far the utilitarian analysis can be taken within this scope.

Virtue ethics raised the most important long-term concern—that repeated AI-assisted workflows may erode the professional judgement that makes meaningful human oversight possible but was the hardest to encode. Deskilling is a dynamic process that unfolds over years of clinical practice. It does not reduce naturally to a binary constraint or a satisfiability check, which meant it remained at the level of analysis rather than implementation [7].

The broader lesson from this project is that ethical goals in AI systems are most effectively addressed when they are treated as first-class design constraints rather than post-hoc commentary. The cases where the implementation fell short particularly around the responsibility gap and the fairness simulation were precisely the cases where the real system's opacity made formal constraint verification difficult. That difficulty is itself an ethical signal.

Acknowledgements

This project drew on the following published works, which informed the system selection, ethical framework construction, and critical analysis presented throughout this report.

The primary clinical foundation was the validation study of the Epic Sepsis Model [1]. The broader ethical, legal, and technical analysis drew on work examining interpretability of clinical AI systems [4], AI integration in healthcare decision support [2], reporting guidelines for AI-driven clinical tools [6], ethical and legal considerations of AI medical decision support [3], explainability in clinical decision support systems [5], and the impact of explanations on advice-taking in AI-based CDSS [7].

Bibliography

1. Cull, J., Brevetta, R., Gerac, J., Kothari, S. & Blackhurst, D. Epic Sepsis Model Inpatient Predictive Analytic Tool: A Validation Study. *Critical Care Explorations* **5**. PMID: 37405252, PMCID: PMC10317482, e0941 (2023).
2. Shwede, F. & Alzoubi, H. M. in *Innovative Dynamics in Management and Engineering* (eds Alzoubi, H. M., Cheng Ling, T. & El Khatib, M.) (Springer, Cham, 2025).
3. Čartolovni, A., Tomičić, A. & Lazić Mosler, E. Ethical, legal, and social considerations of AI-based medical decision-support tools: A scoping review. *International Journal of Medical Informatics*, 104738 (2022).
4. Xu, Q. *et al.* Interpretability of Clinical Decision Support Systems Based on Artificial Intelligence from Technological and Medical Perspective: A Systematic Review. *Journal of Healthcare Engineering* (2023).
5. Amann, J. *et al.* To explain or not to explain? Artificial intelligence explainability in clinical decision support systems. *PLOS Digital Health* **1** (2022).
6. Vasey, B. *et al.* Reporting guideline for the early stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. *BMJ* **377**, e070904 (2022).
7. Panigutti, C., Beretta, A., Giannotti, F. & Pedreschi, D. Understanding the impact of explanations on advice-taking: a user study for AI-based clinical Decision Support Systems in CHI '22: Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (2022).